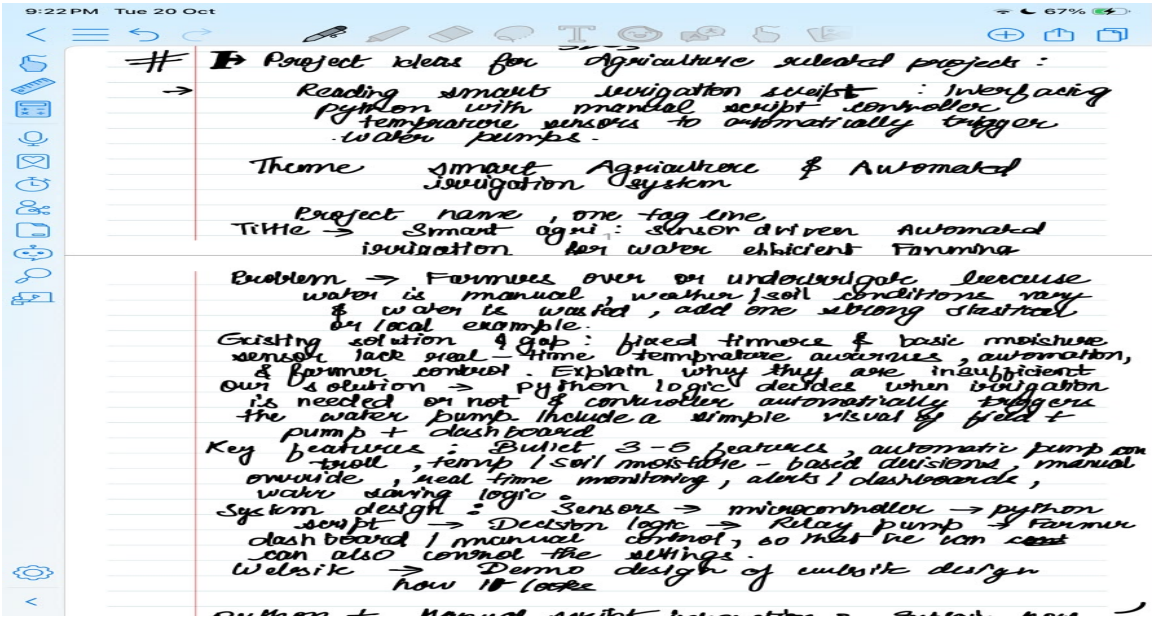
FIELDPULSE

SMART IRRIGATION & WATER-EFFICIENT FARMING

Project concept report

A sensor-driven irrigation system with local decision logic, automatic pump control, manual override and an interactive dashboard concept.



Reference image: handwritten project planning notes supplied by the project team.

Executive summary

FieldPulse is a student project concept for improving irrigation decisions through local sensor readings. A controller reads soil moisture and temperature, applies configurable crop thresholds and safety checks, and issues a pump or valve command when appropriate. A dashboard can display conditions, status and event history. The essential control loop is designed to operate locally, while connectivity can be added for remote monitoring and later synchronization.

The project is presented as a prototype concept. The website simulator is software-only and does not control real hardware. Any water-saving benefit must be measured through controlled field testing rather than assumed from the concept.

1. Problem statement

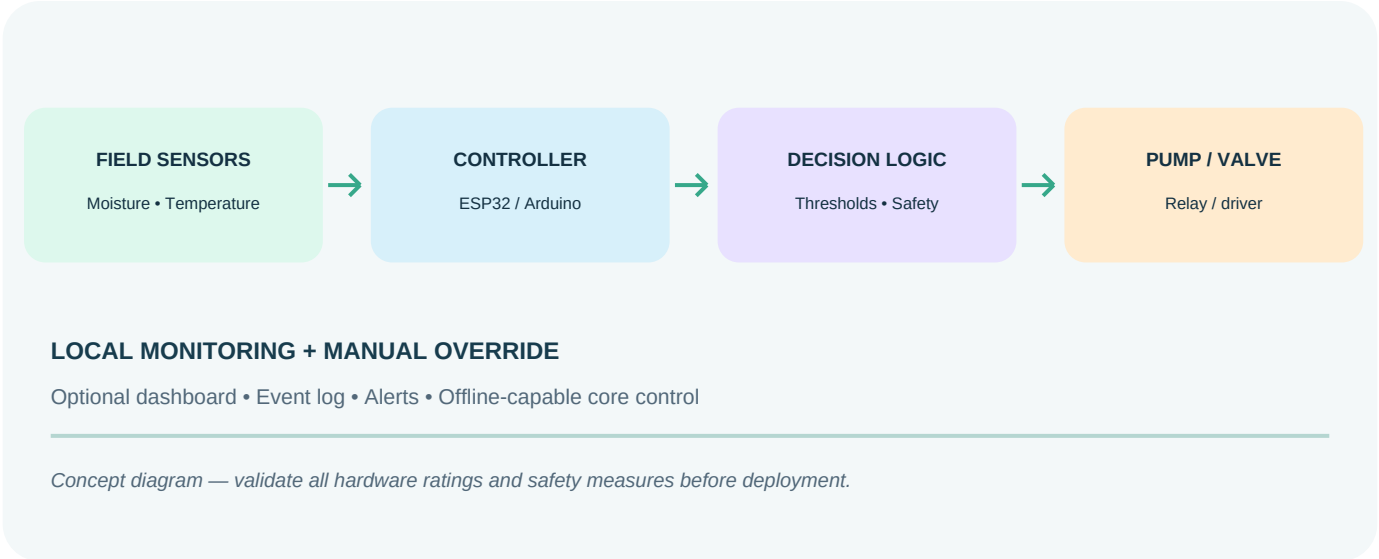
Manual irrigation and fixed schedules can fail to reflect current soil conditions. Farmers may water when the root zone is already sufficiently moist or miss the right time when conditions change. Temperature, rainfall, crop type, soil texture and growth stage all influence irrigation needs. Low-cost sensor-driven automation can help make the process more observable and consistent, but sensor placement and calibration are essential.

2. Proposed solution

A sensor-to-controller-to-actuator architecture collects field data, checks the readings, evaluates crop-specific thresholds, applies safety rules and controls an appropriately rated pump or valve driver. A simple dashboard gives the farmer visibility into readings and allows manual control. The local controller should remain responsible for essential operation even if internet access is unavailable.

3. System architecture

The system can be organized into four layers: sensing, edge controller, decision logic and actuation. A monitoring layer stores events and exposes settings, status and alerts. Optional connectivity can later support remote viewing, multilingual voice guidance, weather integration and offline data synchronization.



Concept diagram — validate all hardware ratings and safety measures before deployment.

System architecture at a glance

Conceptual signal path from field measurements to controlled irrigation action. This illustration is not a wiring diagram.

4. Hardware and software components

Component	Purpose / design notes
Soil moisture sensor	Prefer a capacitive sensor for prototype testing; calibrate for the specific soil and installation depth.
Temperature sensor	Provides environmental context; temperature alone should not determine irrigation.
Optional rain / tank sensors	Can pause irrigation during rain or prevent a pump command when water is unavailable.
Microcontroller	ESP32 or Arduino-compatible board for sensor acquisition and local control. Raspberry Pi may suit a higher-level gateway.
Relay / driver	Must be correctly rated and isolated for the actual pump or valve. Never switch mains loads directly from a microcontroller pin.
Local dashboard	Displays measurements, mode, pump state, thresholds, warnings and event history.
Local data store	Stores timestamped readings and events; SQLite or a lightweight embedded store may be used if a gateway is present.

5. Decision logic

Illustrative flow: read sensors → validate readings → check safety interlocks → compare soil moisture with the selected crop target → consider rain and scheduling rules → issue pump request or keep pump off → record the event. Use hysteresis (a start threshold and a separate stop threshold) to avoid rapid switching around one boundary.

Example rule set

• If the tank is empty or the sensor reading is invalid, block automatic irrigation and raise a warning.
• If rain is detected, pause automatic irrigation according to the configured policy.
• If soil moisture falls below the start threshold and all safety checks pass, request irrigation.
• Stop when the stop threshold is reached, a maximum run time expires, the tank becomes unavailable, or a fault is detected.
• Manual override should be visible, time-bounded where appropriate, and never bypass critical electrical or dry-run safety protections.

The website uses a simplified illustrative threshold model. Its values are not agronomic recommendations and must be calibrated to crop, soil, sensor and field conditions.

6. Dashboard features

• Soil moisture and temperature controls for the demo simulator.
• Crop profile selection with adjustable example thresholds.
• Automatic and manual modes, rain and tank interlocks.
• Pump request state, moisture gauge, status explanations and an event log.
• Searchable sections, keyboard shortcuts, guided tour, dark/light mode and responsive layout.
• Project overview, architecture, build plan, impact considerations, roadmap, FAQs and contributor cards.

7. Testing and evaluation plan

Bench test sensor readings against a reference measurement. Record calibration curves and identify noise, drift and out-of-range values. Test each safety interlock independently. Verify that the pump output defaults to a safe state during controller reset, power loss or sensor failure. Before any field trial, use an appropriate isolated driver and a supervised low-risk test setup.

For impact evaluation, compare a baseline irrigation practice with the prototype over similar periods. Track water volume, irrigation duration, soil moisture, rainfall, crop stage and manual interventions. Report actual results with conditions and limitations; do not claim guaranteed savings from the simulator.

8. Safety, reliability and limitations

This is an educational concept and front-end simulation. It is not a certified irrigation controller, does not make live measurements, and is not connected to a physical pump. A real implementation requires electrical design review, correctly rated and isolated switching hardware, waterproof enclosures, protection against dry running and overcurrent, secure wiring, fail-safe defaults and supervised testing.

Sensors require calibration and can be affected by soil composition, salinity, installation depth, corrosion and local conditions. Thresholds vary by crop, growth stage and soil type. A single sensor may not represent an entire field; multiple sensor locations may be necessary for larger plots.

9. Future scope

Phase 1 — interactive software concept and project documentation.
Phase 2 — bench prototype with calibrated sensors and safe low-voltage output testing.
Phase 3 — supervised field pilot with measured water use and soil observations.
Phase 4 — optional remote monitoring, offline synchronization, multilingual voice prompts, weather-aware planning and anomaly detection. Any AI or predictive feature should be validated against measured field data before operational use.

10. Project contributors

Aryan Das — contributor; Lovely Professional University. Profile links supplied for the project: LinkedIn · GitHub. The profile summary on the website is a project-focused draft and should be checked against the current public profile before publication.

Asmi Jha — project contributor. Profile link supplied for the project: LinkedIn. A verified professional summary and GitHub URL can be added when confirmed by the contributor.

11. Conclusion

FieldPulse demonstrates how an affordable, sensor-driven system can make irrigation status more visible and automate a basic decision loop while preserving farmer control. The strongest next step is a safe, measurable prototype: calibrate sensors, implement robust interlocks, record events, and evaluate performance in a controlled test before drawing conclusions about water savings or crop outcomes.

Prepared as a student project concept report. All thresholds and dashboard readings in the web simulator are illustrative demo values.